Cell Biology
Cancer Biology

Human *RAP2A* Homolog of the *Drosophila* Asymmetric Cell Division Regulator *Rap2l* Targets the Stemness of Glioblastoma Stem Cells

Reviewed Preprint
v2 • October 31, 2025
Revised by authors

Reviewed Preprint
v1 • May 1, 2025

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eLife Assessment

This **useful** study explores the role of RAP2A in asymmetric cell division (ACD) regulation in glioblastoma stem cells (GSCs), drawing parallels to Drosophila ACD mechanisms and proposing that an imbalance toward symmetric divisions drives tumor progression. While findings on RAP2A's role in GSC expansion are promising, and the reviewers found the study innovative and technically sound, the study is nevertheless still considered **incomplete** because of its reliance on neurosphere models without in vivo confirmation and insufficient mechanistic validation. Addressing those gaps would substantiate the study's claims.

https://doi.org/10.7554/eLife.105690.2.sa3

Abstract

Asymmetric cell division (ACD) is a fundamental process to balance cell proliferation and differentiation during development and in the adult. Cancer stem cells (CSCs), a very small but highly malignant population within many human tumors, are able to provide differentiated progeny by ACD that contribute to the intratumoral heterogeneity, as well as to proliferate without control by symmetric, self-renewing divisions. Thus, ACD dysregulation in CSCs could trigger cancer progression. Here we consistently find low expression levels of *RAP2A*, the human homolog of the *Drosophila* ACD regulator *Rap2l*, in glioblastoma (GBM) patient samples, and observe that scarce levels of *RAP2A* are associated with poor clinical prognosis in GBM. Additionally, we show that restitution of RAP2A in GBM neurosphere cultures increases the ACD of glioblastoma stem cells (GSCs), decreasing their proliferation and expression of stem cell markers. Our results support that ACD failures in GSCs increases their spread, and that ACD amendment could contribute to reduce the expansion of GBM.

Introduction

Asymmetric cell division (ACD) is a universal mechanism for generating cellular diversity during development as well as for regulating tissue homeostasis in the adult (1). Stem and progenitor cells undergo ACD to simultaneously give rise to a self-renewing stem/progenitor cell and to a daughter cell that is committed to enter a differentiation program. Remarkably, over the past years it has become apparent that undermining this critical process can lead to tumor-like overgrowth (2). This link between failures in the process of ACD and tumorigenesis was first established using the *Drosophila* neural stem cells, called neuroblasts (NBs), as a model system (3). In this work, authors showed how *Drosophila* larval mutant brains for ACD regulatory genes were able to induce the formation of massive tumoral masses after weeks of being transplanted into the abdomen of wild-type fly hosts. Intriguingly, some genes firstly identified in *Drosophila* as tumor suppressor genes, such as *discs large 1* (*dlg1*), *lethal (2) giant larvae* (*l(2)gl*) and *brain tumor* (*brat*) (4), were later uncovered as ACD regulators (5-11), further supporting the connection between compromised ACD and tumorigenesis.

Tumor initiating cells or cancer stem cells (CSCs) were originally identified in acute myeloid leukemia (AML) (12,13) but, to date, several studies have also revealed the existence of CSCs in multiple solid tumors (14). These CSCs represent a very small but highly malignant population within the tumor, able to proliferate without control by symmetrical, self-renewal divisions, as well as to provide differentiated progeny by ACD that contribute to the heterogeneity of the tumor. These properties, along with their quiescent state, increase in metabolic activity and high capacity of DNA repair, make CSCs extremely resistant to radio- and chemotherapy and responsible for malignant relapse (14-18). Thus, understanding the unique nature of these cells and their landmarks is crucial to target CSCs and hence to completely destroy the tumor. An intriguing possibility is that an imbalance in the mode of CSC divisions, favoring symmetric renewal divisions in detriment of asymmetric divisions, might contribute to the transition from a chronic (low grade) to an acute (high grade) phase in the cancer progression (19,20).

Glioblastoma (GBM), the most aggressive and lethal brain tumor with very poor prognosis, is among the solid tumors in which the presence of CSCs, called glioma or glioblastoma stem cells (GSCs), has been proven. As other CSCs, GSCs are responsible for the formation, maintenance, resistance to conventional therapy and consequent relapse of GBM (21-25). Several GSC biomarkers, such as CD133 and Nestin, and signaling pathways, such as Notch and Hedgehog, have been identified to help isolate and target these crucial cells (14,26,27). Despite of that, there are still many unknowns regarding the nature, properties and origin of CSCs in general and of GSCs in particular.

Here, we analyze the consequences of compromising ACD in the GSCs within GBM neurosphere cell cultures. Specifically, we show that *RAP2A*, the human homolog of *Rap2l*, which encodes a *Drosophila* small GTPase and novel ACD regulator, displays low expression levels in GBM, and that restitution of RAP2A in GBM neurosphere cultures increases the ACD and decreases the expression of stem cell markers in the GSCs. Our results support that failures in ACD increase GSC proliferation promoting their spreading and that ACD amendment might contribute to reduce the expansion of GBM.

Results

RAP2A is highly downregulated in GBM patients

As a first approach to analyze whether ACD might be impaired in GBM, we analyzed the expression levels of human homologs of 21 *Drosophila* ACD regulators in a GBM microarray (28) (**Fig. 1A**). We also investigated which of those genes were more similar in expression pattern to the others by performing a Gene Distance Matrix (**Fig. 1B**). The human genes *TRIM2* and *RAP2A* showed the lowest levels in the GBM patient samples analyzed. *TRIM2* belongs to the same family that *TRIM3* and *TRIM32*, all of them related to the *Drosophila* ACD regulator gene *brat*. *TRIM3*, the closest homolog of *brat*, has been previously found to be expressed at low levels in GBM (29). Hence, we concentrated on *RAP2A*, the second human gene consistently found at very low levels in all the patient samples (**Fig. 1A** and **Fig. S1**). *RAP2A* is also known to have a tumor suppressor role in the context of glioma migration and invasion (30,31). First, to further support the microarray data expression levels in a bigger sample size, we performed *in silico* analyses taking advantage of established datasets of Glioblastoma IDH wild-type, such as "The Cancer Genome Atlas" (TCGA) and "Gravendeel" cohorts. We confirmed that low *RAP2A* expression levels in human GBM are associated with a poor prognosis (**Fig. 1C**). In a multivariate analysis in the TCGA cohort, when we included the variables RAP2A expression, patient age and standard chemo/radiotherapy treatment in the model, it was observed that RAP2A levels and patient age have a significant implication in patient survival as an independent variable. In fact, specific values were significant in predicting overall survival (OS). For example, "RAP2A expression" with a hazard ratio (HR) of 0.866 (95% CI 0.760-0.986) and a p-value of 0.03, and "patient age" with a hazard ratio (HR) of 1.030 (95% CI 1.022-1.037) and a p-value of <0.001. Conversely, standard chemo/radiotherapy treatment did not remain as an independent variable, with a hazard ratio (HR) of 0.820 and a p-value of 0.053. We also analyzed RAP2A levels in the different GBM subtypes (proneural, mesenchymal and classical) in the TCGA cohort and their prognostic relevance. The proneural subtype that displayed RAP2A levels significantly higher than the others was the subtype that also showed better prognosis (**Fig. S2**). Hence, we focused on the human gene *RAP2A* to analyze its potential role in ACD and its relevance in GBM.

Drosophila RAP2A homolog Rap2l regulates ACD

We previously showed that the *Drosophila* gene *Rap1* regulates the ACD process (32). The closest human homolog of *Drosophila Rap1* is *RAP1A*, while human *RAP2A* is more similar to *Drosophila Rap2l*, which has not been previously analyzed in the context of ACD. Hence, we first wanted to investigate whether *Rap2l* was, like *Rap1*, implicated in ACD regulation. With that aim, we started analyzing the ACD of neural stem cells (neuroblasts, NBs) and intermediate neural progenitors (INPs) within *Drosophila* larval brain type II NB lineages (NBII) (**Fig. 2A**) (8,33,34). In NBII lineages, there is one NB that expresses the transcription factor Deadpan (Dpn) and several INPs, which express both Dpn and the transcription factor Asense (Ase) (**Fig. 2A**). We first observed the presence of ectopic NBs (eNBs; Dpn^+ Ase^-) after downregulating *Rap2l* in NBII lineages in 4 out of 5 mutant brains analyzed (n = 34 NB lineages), while in control lineages only one NB was usually found (**Fig. 2B**). To further determine the specificity and penetrance of the phenotype, we repeated this experiment analyzing this and two additional *Rap2l* RNAi lines, substantially increasing the number of the samples (both the number of NB lineages and the number of brains analyzed) (**Fig. S3**). All the different lines showed a highly significant number of ectopic NBs (**** p < 0.001; n> 120 NB lineages analyzed) in all or almost all of the different brains analyzed (n>14) (see **Fig. S3** for details). This result suggested that the process of ACD was impaired within those *Rap2l* mutant NBII lineages. To more directly support that, we analyzed the localization of key ACD regulators in metaphase NBs and INPs of NBII lineages, such as the apical proteins aPKC and Canoe (Cno), and the cell-fate determinant Numb (35-37). In control metaphase NBs and INPs, those ACD regulators form crescents at the apical (aPKC and Cno) or basal (Numb) poles of the dividing cell (**Fig. 2C**). The apical localization of aPKC was not significantly altered after downregulating *Rap2l* in NBII lineages; however, the localization of Cno and Numb did fail in most of the *Rap2l* mutant brains analyzed (in 5 out of 6 in the case of Cno,

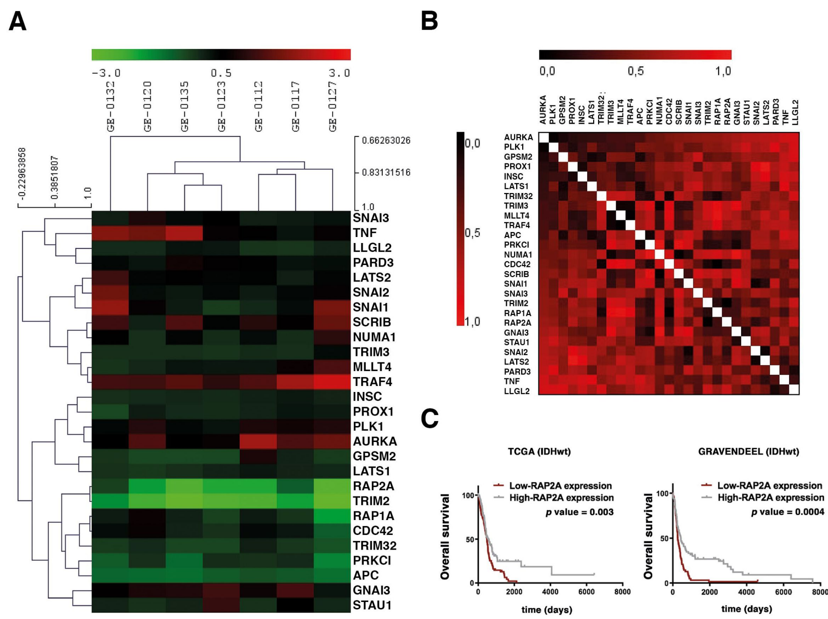


Figure 1.

RAP2A is highly downregulated in GBM patients.

(**A, B**) Analysis of GBM patient versus control samples focused on the level of expression of human homologs of *Drosophila* ACD regulators by Hierarchical clustering (**A**) and the Gene Distance Matrix (**B**). Color-coded scale bar indicates fold level of expression. **(C)** Kaplan-Meier survival curves corresponding to patients in the TCGA (IDHwt) and Gravendeel (IDHwt) cohort in GBM datasets. Low *RAP2A* expression levels in human GBM are associated with a poor prognosis.

and in 3 out of 4 in the case of Numb) (**Fig. 2C**). All these results strongly suggested that Rap2l, as Rap1, was an ACD regulator. Hence, we came back to the human homolog of *Drosophila Rap2l*, *RAP2A*, to determine its potential relevance in the context of the ACD of GSCs.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population

After confirming by *in silico* analyses that low *RAP2A* expression levels in GBM patients were associated to a poor prognosis, we decided to use established GBM cell lines to investigate in neurosphere cell cultures whether the downregulation of *RAP2A* was affecting the properties of GSCs. We tested six different GBM cell lines finding similar mRNA levels of *RAP2A* in all of them, and significantly lower levels than in control Astros (**Fig. 3A**). We focused on the GBM cell line called GB5, which grew well in neurosphere cell culture conditions, for further analyses. Given that stem cell markers, such as CD133, SOX2 and NESTIN are landmarks of GSCs, we started analyzing their expression after restoring RAP2A in the neurosphere culture (**Fig. 3B**). Immunofluorescences of each of those markers revealed a significant reduction in the protein intensity in the RAP2A-expressing neurospheres (GB5-RAP2A) compared to the control neurospheres (GB5) (**Fig. 3C**). Likewise, RT-PCRs and Westerns blots also showed a significant decrease in the mRNA and protein expression levels, respectively, of those stem cell markers (**Fig. 4**). Hence, RAP2A seems to be contributing to decrease the stem cell population within GBM neurosphere cultures.

RAP2A expression in GBM neurosphere cultures decreases cell proliferation and sphere size

Next, we wondered whether the reduction in the stem cell population after expressing *RAP2A* in neurosphere cultures was reflected in the level of cell proliferation and, consequently, in the size of the neurospheres. To investigate this, we first analyzed the expression of the proliferation marker and key tool in cancer diagnostics Ki-67, highly expressed in cycling cells but absent in quiescent (G0) cells (38,39). We detected a significant decrease in the number of proliferating cells per neurosphere in those cultures expressing *RAP2A* (**Fig. 5A**). Moreover, the average area size of these GB5-RAP2A neurospheres was also significantly reduced compared to control GB5 neurospheres (**Fig. 5B**). Thus, RAP2A promotes a decrease in cell proliferation.

RAP2A expression in GBM neurosphere cultures fosters ACD in GSCs

Given the reduction in cell proliferation and in the stem cell population after adding RAP2A to the control GB5 neurosphere cultures, we wanted to investigate whether RAP2A was favoring an asymmetric mode of cell division in these cells. Thus, we decided to follow over time the GB5-RAP2A and the control GB5 neurospheres, analyzing the number of cells per neurosphere. We reasoned that an even number of cells per neurosphere would be indicative of symmetric cell divisions, while an odd number of cells would point to an asymmetric mode of cell division (see also Material and methods for details). We observed a significant increase of neurospheres with an odd number of cells in the GB5-RAP2A neurospheres (n= 124 neurospheres analyzed) compared to control GB5 neurospheres (n=153 neurospheres) (**Fig. 6A**). This result was very suggestive of an increase in the number of ACDs in GB5-RAP2A neurosphere cells. To further support this result, we decided to analyze the distribution of the cell fate determinant NUMB, a key conserved ACD regulator whose asymmetric distribution in the mother and progeny ultimately promotes the repression of stem cell self-renewal in the daughter cell in which is segregated (40-44). We

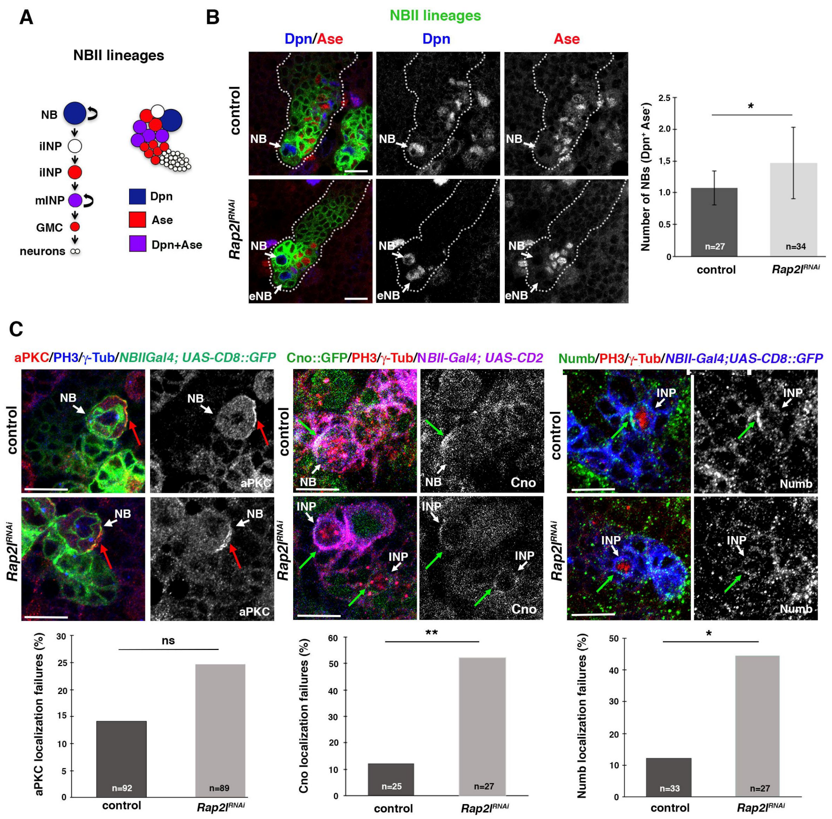


Figure 2.

Drosophila RAP2A homolog *Rap2l* regulates ACD.

(**A**) *Drosophila* NBII lineage; the only NB per lineage expresses the transcription factor Dpn, while mature INPs express both Dpn and Ase; iINP, immature INP; mINP, mature INP; GMC, Ganglion Mother Cell. **(B)** Confocal micrographs showing a control larval brain NBII lineage with one NB (Dpn^{+} Ase^{-}) and an NBII lineage in which *Rap2l* has been downregulated displaying an ectopic NB (eNB). **(C)** *Rap2l* downregulation in NBII lineages by the *wor-Gal4 ase-Gal80* (an NBII-specific driver) causes Cno and Numb localization failures (green arrows) in dividing progenitors (NB or INPs), while aPKC localization is not significantly altered (red arrows). Data shown in the scaled bar graphs was analyzed with a U Mann-Whitney test (B) or a Chi-square test (C), **p*<0.05, ***p*<0.01, ns, not significant, n=number of NB lineages analyzed (in B) or number of dividing cells analyzed (in C); scale bar: 10 µm.

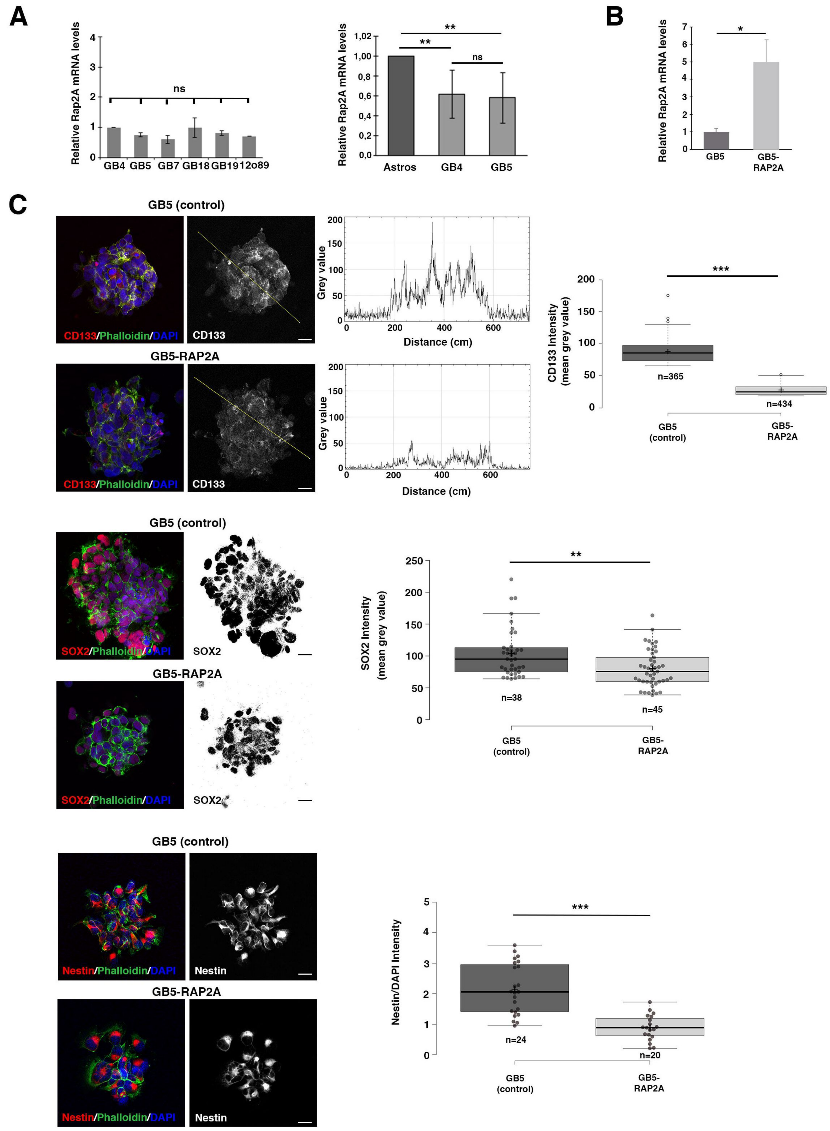


Figure 3.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population.

(A) Different GBM cell lines show similar *RAP2A* mRNA levels and significantly lower levels than in control Astros. **(B)** *RAP2A* mRNA levels are significantly higher in the GB5 line after infecting this line with *RAP2A* (GB5-RAP2A). Data shown in the scaled bar graphs in A and B was analyzed with an ANOVA and a T-test, respectively; error bars show the SD; n= 2 (in A) and 3 (in B) different experiments. **(C)** Immunofluorescences of the GSC stem cell markers CD133, SOX2 and Nestin reveal a significant reduction in the protein intensity in the GB5 RAP2A-expressing neurospheres (GB5-RAP2A) compared to the control neurospheres (GB5). Data shown in the box plots was analyzed with a U Mann-Whitney for CD133 and Nestin and with a T-test for Sox; the central lines represent the median and the box limits the lower and upper quartiles, as determined by R software; crosses represent sample means; error bars indicate the SEM; n= number of sample points; *p* values: **p*<0.05, ***p*<0.01, ****p*<0.001, ns, not significant; scale bar: 10 µm.

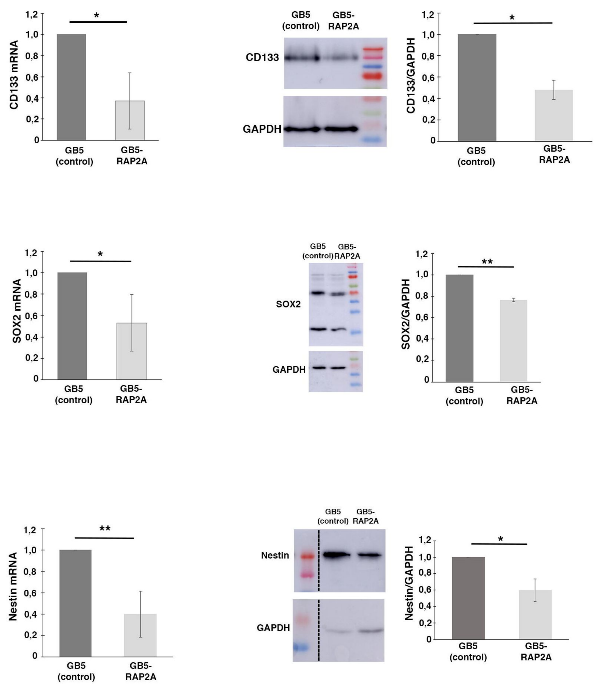


Figure 4.

RAP2A expression in GBM neurosphere cultures reduces the stem cell population.

RT-PCRs and Westerns blots show a significant decrease in the mRNA and protein expression levels, respectively, of the GSC markers CD133, SOX2 and Nestin in the GB5 RAP2A-expressing neurospheres (GB5-RAP2A) compared to the control neurospheres (GB5). Data shown in the scaled bar graphs was analyzed with an unpaired two-tailed Student's *t* test; error bars show the SD; n= 3 different experiments; *p* values: *p<0.05, **p<0.01.

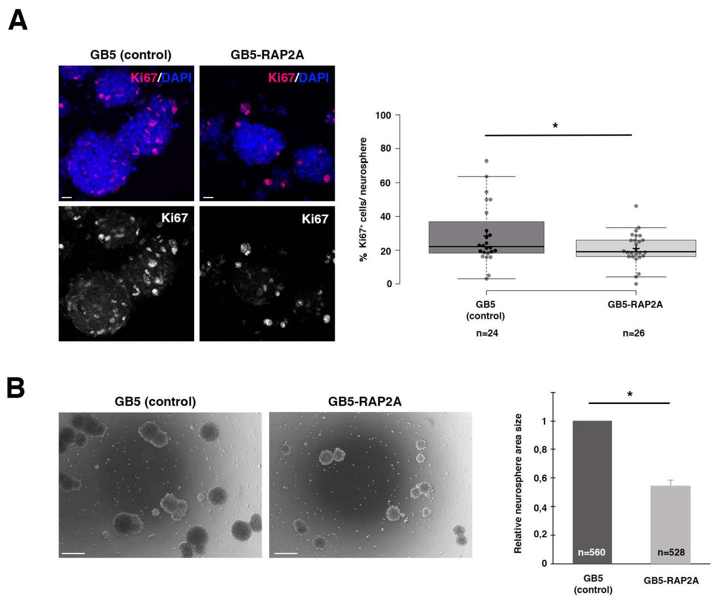


Figure 5.

RAP2A expression in GBM neurosphere cultures decreases cell proliferation and sphere size.

(A) GB5 neurospheres expressing RAP2A (GB5-RAP2A) show a significantly lower number of Ki67 expressing cells per neurosphere than control GB5 neurospheres. Data shown in the box plots was analyzed with a T-test; the central lines represent the median and the box limits the lower and upper quartiles, as determined by R software; crosses represent sample means; error bars indicate the SEM; n= number of sample points. **(B)** GB5 neurospheres expressing RAP2A (GB5-RAP2A) show a significant decrease in their size compared to control GB5 neurospheres of the same stage. Data shown in the scaled bar graphs was analyzed with an unpaired two-tailed Student's T-test; error bars show the SD; n= total number of neurospheres of 3 different experiments; *p* values: *$p<0.05$; scale bars: 10 µm (in A) and 100µm (in B).

observed a significant increase in the number of GB5-RAP2A dividing cells that showed an asymmetric localization of NUMB in the progeny, compared to control GB5 dividing cells (**Fig. 6B**). Thus, all these results strongly supported a function of RAP2A promoting ACD in the GSCs.

Discussion

ACD is an evolutionary conserved mechanism used by stem and progenitor cells to generate cell diversity during development and to regulate tissue homeostasis in the adult. CSCs, present in many human tumors, can divide asymmetrically to generate intratumoral heterogeneity or symmetrically to expand the tumor by self-renewal. Over the past 15 years, it has been suggested that dysregulation of the balance between symmetric and asymmetric cell divisions in CSCs, favoring symmetric divisions, can trigger tumor progression in different types of cancer (19,20,45) including mammary tumors (43,46), glioblastoma (29), oligodendrogliomas (47,48), colorectal cancer (49,50) and hepatocellular carcinoma (50). Thus, it is of great relevance to get a deeper insight into the network of regulators that control ACD, as well as the mechanisms by which they operate in this key process.

Drosophila neural stem cells or NBs have been used as a paradigm for many decades to study ACD (51). NBs divide asymmetrically to give rise to another self-renewing NB and a daughter cell that will start a differentiation process. Over all these years, a complex network of ACD regulators that tightly modulate this process has been characterized. For example, the so-called cell-fate determinants, including the Notch inhibitor Numb, accumulate asymmetrically at the basal pole of mitotic NBs and are exclusively segregated to one daughter cell, promoting in this cell a differentiation process. The asymmetric distribution of cell-fate determinants in the NB is, in turn, regulated by an intricate group of proteins asymmetrically located at the apical pole of mitotic NBs, generically known as the “apical complex”. This apical complex includes kinases (i.e. aPKC), small GTPases (i.e. Cdc42, Rap1) and Par proteins (i.e. Par-6, Par3), among others (51).

Given the potential relevance of ACD in CSCs, we decided to take advantage of all the knowledge accumulated in *Drosophila* about the network of modulators that control asymmetric NB division. As a first approach, we aimed to analyze whether the levels of human homologs of known *Drosophila* ACD regulators were altered in human tumors. Specifically, we centered on human GBM, as the presence of CSCs (GSCs) have been shown in this tumor. The microarray we interrogated with GBM patient samples had some limitations. For example, not all the human genes homologs of the *Drosophila* ACD regulators were present (i.e. the human homologs of the determinant Numb). Likewise, we only tested seven different GBM patient samples. Nevertheless, the output from this analysis was enough to determine that most of the human genes tested in the array presented altered levels of expression. We selected for further analyses *RAP2A*, one of the human genes that showed the lowest levels of expression, compared to the control samples. However, it would be interesting to analyze in the future the potential consequences that altered levels of expression of the other human homologs in the array can have in the behavior of the GSCs. In silico analyses, taking advantage of the existence of established datasets, such as the TCGA, can help to more robustly assess, in a bigger sample size, the relevance of those human genes expression levels in GBM progression, as we observed for the gene *RAP2A*.

We have previously shown that *Drosophil*a Rap1 acts as a novel NB ACD regulator in a complex with other small GTPases and the apical regulators Canoe (Cno), aPKC and Par-6 (32). Here we have shown that *Drosophila* Rap2l also regulates NB ACD by ensuring the correct localization of the ACD modulators Cno and Numb. We also have demonstrated that RAP2A, the human homolog of *Drosophila* Rap2l, behaves as an ACD regulator in GBM neurosphere cultures, and its restitution to these GBM cultures, in which is present at low levels, targets the stemness of GSCs increasing the number of ACDs. It would be of great interest in the future to determine the specific mechanism by which Rap2l/RAP2A is regulating this process. One possibility is that, as it occurs in

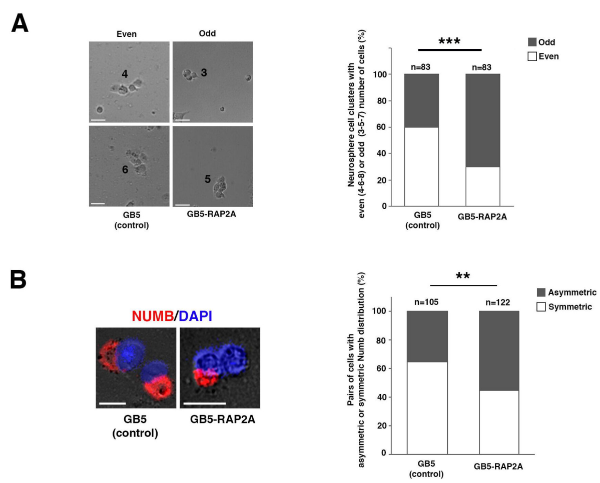


Figure 6.

RAP2A expression in GBM neurosphere cultures fosters ACD in GSCs.

(A) Early stage GB5 neurospheres expressing RAP2A (GB5-RAP2A) show an odd number of cells (3-5-7) significantly more frequently than control GB5 neurospheres, which show more frequently an even number of cells (4-6-8). **(B)** GB5-RAP2A dividing cells show a significant increase in the number of asymmetric NUMB localization in the progeny, compared to control GB5 dividing cells. Data shown in the bar graphs was analyzed with a Chi-Square test with Yates correction; n= number of neurosphere cell clusters analyzed. *p* values: ***p*<0.01; ****p*<0.001scale bar: 20 µm.

the case of the *Drosophila* ACD regulator Rap1, Rap2l/RAP2A is physically interacting or in a complex with other relevant ACD modulators. Thus, this study supports the relevance of ACD in CSCs of human tumors to refrain the expansion of the tumor. Other studies, however, claim that ACD should be targeted in human tumors as it promotes the intratumoral heterogeneity that hampers the complete tumor loss after chemotherapy (52-54). More investigations should be carried out with other human genes homologs of ACD regulators to further confirm the results of this study. Likewise, analyses in vivo (i.e. in mouse xenografts) would be also required to reinforce our conclusions. This would be very relevant in order to consider ACD restitution in CSCs of human tumors, what has been called "differentiation therapy" (55), as a potential alternative therapeutic treatment.

Materials and Methods

Drosophila husbandry, strains and genetics

The following fly stocks were used: *wor-Gal4 ase-Gal80 (56)*; *UAS-CD8::GFP* (Bloomington *Drosophila* Stock Center -BDSC-#5137); *UAS-CD2* (BDSC #1373); *UAS-Rap2l*^{KKRNAi} (Vienna *Drosophila* Resource Center -VDRC-#107745); *UAS-Rap2l*^{GDRNAi} (VDRC #45228); *UAS-Rap2l*^{RNAi} (BDSC #51840); *cno::GFP (57)*. All the fly stocks were raised and kept in 18°C or 25°C incubators. Experimental temperatures for the assays were maintained using 25°C or 29°C incubators. The *Gal-4* x *UAS*-crosses were carried out at 29°C (the first 2 days they were kept at 25°C, and then transferred to a new tube and left at 29°C) until third instar larvae developed.

Drosophila histology, immunofluorescence and microscopy

Larval brains were dissected out in PBS and fixed with 4% PFA in PBT (PBS and Triton X-100 0.1%) for 20 min at room temperature with gentle rocking. Fixed brains were washed 3 times for 15 min with PBT (PBS and Triton X-100 0.3%) and then incubated in PBT-BSA for at least 1h before incubation with the corresponding primary antibody/antibodies. The following primary antibodies were used in this study: guinea pig anti-Dpn (1:2,000) (57), rabbit anti-Ase (1:100) (57), rabbit anti PKCζ (1:100; Santa Cruz Biotechnology, sc-216), goat anti-Numb (1:200; Santa Cruz Biotechnology, sc-23579), mouse anti-PH3 (1:2,000; Millipore, 05-806), mouse anti-γTub (1:200; Sigma-Aldrich, T5326) and mouse anti-CD2 (1:50; BioRad). Secondary antibodies conjugated to fluorescent dyes (Invitrogen) were used at 1:400 dilution. Samples were mounted in VECTASHIELD antifade mounting medium for fluorescence (Vector Laboratories, H-1000). Fluorescent images were captured using a Super-resolution Inverted Confocal Microscope Zeiss LSM 880-Airyscan Elyra PS.1. Images were analyzed using the image processing package FIJI from ImageJ and assembled using Adobe Photoshop CS6.

GBM microarray

The cohort of patients consisted of seven GBM biopsies obtained from the Valencian Network of Biobanks. The procedures for obtaining tissue samples were developed in accordance with national ethical and legal standards, and following the guidelines established in the Declaration of Helsinki, and approved of Clinical Research Ethics Committee of the Hospital General Universitario de Elche, Spain. All patients had been diagnosed with GBM grade IV, and all of them exhibited an *IDH1* plus *IDH2* wild-type genotype. Normal human brain RNA (frontal lobe; pool from 5 donors) was purchased from BioChain and used as a control for gene expression experiments. A total of 12,516 genes were used to generate a custom Agilent Two-Color 8 x 15 K Agilent gene expression DNA microarray, which were deposited in the NCBI Gene Expression Omnibus (GEO) Database under accession number GSE182697 (28).

Bioinformatics analysis

The level of expression of human homologs of *Drosophila* ACD regulators was assessed in seven different GBM patients versus control samples of the custom-made microarray (GSE182697) (28). Statistical analysis of microarray data was done in Multiexperiment Viewer (MeV) (58). Hierarchical clustering analyses were done optimizing gene and sample leaf order and using Pearson Correlation for the distance metric selection. Average linkage clustering was used as a linkage method. For the Gene Distance Matrix (GDM), the distance, inverse of similarity, between two genes was calculated using a distance metric as indicated in MeV.

Human samples and cell lines

Glioma samples were obtained with patients' written informed consent, in accordance with the Declaration of Helsinki, and approved by the Ethics Committee at Hospital 12 de Octubre, Madrid, Spain (CEI 14/023). The primary glioblastoma cell lines were provided by the Biobank at "Hospital 12 de Octubre". Fresh tissue samples were enzymatically dissociated using Accumax (Millipore) and cultured in stem cell medium (Neurobasal, Invitrogen), supplemented with N2 (1:100, Invitrogen), GlutaMAX (1:100, Invitrogen), penicillin-streptomycin (1:100, Lonza), 0.4% heparin (Sigma-Aldrich), 40 ng/ml EGF and 20 ng/ml bFGF2 (Peprotech). The GB cell lines used in this study were: GB4, GB5, GB7, GB18, GB19, 12o89, Astros (59,60) and HEK293T. The cell lines HEK293T and Astros were maintained in Dulbecco's modified Eagle Medium supplemented with 10% fetal bovine serum 2 mM L-glutamine, 0.1% penicillin (100 U/ml) and streptomycin (100 µg/ml) and Non-Essential Aminoacids (NEA 1x).

GBM-derived neurosphere cell cultures

For stem cell culture of GBM-derived neuroespheres we used DMEM:F12 with FGF-2 (20 ng/ml), EGF-2 (20 ng/ml), and N2 supplements, 0.1% penicillin (100 U/ml) and streptomycin (100 µg/ml), as previously described (61,62). The tumor spheres were dissociated to single cells using Accumax, and 1×10^5 dissociated cells/ml were plated in a Flask and cultured for 6-7 days.

In Silico Analysis

Data from The Cancer Genome Atlas (TCGA) and the Chinese Glioma Genome Atlas (CGGA) for GBM cohorts were accessed through UCSC Xena-Browser (*https://xenabrowser.net*), and Gliovis (*http://gliovis.bioinfo.cnio.es*) for extraction of overall survival data, gene expression levels, and distribution of genetic alterations. Kaplan-Meier survival curves were generated by stratifying samples into low- and high-expression groups for each gene. Survival differences between these groups were assessed using the log-rank test.

DNA Constructs and Lentiviral Production

Lentiviral vectors were used to produce cells overexpressing *RAP2A* (*pLJM1RAP2A* #Plasmid 19311, Addgene) or eGFP (pLJMEGFP #Plasmid 19319, Addgene) as infection control. Infected cells were selected with Puromycin. To obtain the lentivirus, HEK293T cells were transiently co-transfected with 5 µg of the appropriate lentiviral plasmid (pLJEGFP or pLJRRap2A, respectively), 5 µg of the packaging plasmid pCMVdR8.74 (#Plasmid 22036, Addgene), and 5 µg of VERSUSV-G envelope protein plasmid pMD2G (#Plasmid 12259, Addgene), using Lipofectamine Plus reagent (Invitrogen, Carisbad, CA, USA). Lentiviral supernatants were collected after 48 h of HEK293T transfection.

Neurosphere Immunofluorescences and Imaging

For immunostaining, floating cultured neurospheres were incubated on Matrigel coated micro slide glass for 2 hours at 37oC. After having been washed, the attached neurospheres were fixed in 4% paraformaldehyde for 20 min. Cells were blocked for 1 h in 1% FBS and 0.1% Triton X-100 in PBS; then, they were incubated for 2 hours with the primary antibody: mouse anti-CD133 (HB#7, Developmental Studies Hybridoma Bank), mouse anti-Ki67 (3E6, Developmental Studies Hybridoma Bank), rabbit anti-Sox2 (N1C3, GTX101507 GeneTex), mouse anti-Nestin (10c2, sc-23927, Santa Cruz Biotechnology) and mouse anti-Numb (48, sc-136554, Santa Cruz Biotechnology). The secondary antibodies used were Alexa555 goat anti-rabbit (A32732, Invitrogen) or Alexa555 goat anti-mouse (A32727, Invitrogen). Phalloidin 633 (Sigma-Aldrich, 68825 Phalloidin-Atto 633) was added and incubated for 20 min in PBS; Vectashield Mounting medium with DAPI (Linaris, H-1200) was added before mounting. Fluorescent images were captured using a Super-resolution Inverted Confocal Microscope Zeiss LSM 880-Airyscan Elyra PS.1. Images were analyzed using the image processing package FIJI from ImageJ and assembled using Adobe Photoshop CS6 program.

Immunofluorescent quantification

Fluorescence intensity measurements were carried out using ImageJ software package. In all cases, images of neuroespheres transfected with GFP (GB5, control) or RAP2A (GB5-RAP2A) lentivirus were analyzed:

<u>CD133:</u> Using the "line tool", a line was drawn on the channel with the CD133 signal on three different pictures per genotype. Fluorescence was measured across the line sections with the "Plot Profile Tool", and reported as a subset of images showing the signal intensity plot. For quantification, only the 100 highest gray value points, were taken.

<u>Sox:</u> After adjusting the threshold on the channel with the Sox2 signal and eliminating particles smaller than 5 microns, the fluorescence intensity was measured as the mean gray value (sum of all pixels with gray values inside the selection, divided by the number of pixels). In **Figure 3** is shown one example out of three quantifications achieved with at least 20 cells per genotype.

<u>Nestin:</u> The regions of interest (ROIs) were set with the "hand drawing tool" using the Phalloidin staining to follow the cell periphery. At least 20 cells (ROIs) from three different pictures were selected per condition. The background fluorescence was subtracted from the mean gray value fluorescence for each well, and the fluorescence intensity was normalized by the nuclear DAPI signal coming from the same cell (ROI). Plot in **Figure 3C** represents the mean gray value that is the sum of all pixels with gray values inside the selection divided by the number of pixels.

RNA isolation and qRT-PCR assay

Total RNA was extracted from 7 days old neurospheres using TRI reagent (AM9738, Invitrogen) following manufacturer indications and quantified using a nanodrop (ND-1000, Thermo Scientific). 2ug of RNA was treated with DNaseI (EN0521, Thermo Scientific) and reverse transcribed with SuperScriptTM III Reverse Transcriptase (1808044, Invitrogen). Quantitative real time PCR (qRT-PCR) was performed using Power SYBR Green PCR Master Mix (PN4367218, Applied Biosystems), following established protocols with 60°C for annealing/extension and 40 cycles of amplification on a QuantStudioTM 3 apparatus (applied Biosystems). Act5 primers were used for normalization, and relative expression was calculated using the comparative ΔΔCt. The average of at least 3 independent experiments is shown in **Figure 4**.

Neurosphere Western Blots

Protein extracts were prepared by re-suspending cell pellets in a lysis buffer (50-mM Tris, pH 7.5, 300-mM NaCl, 0.5% SDS, and 1% Triton X-100) and incubating the cells for 10 min at 95oC. The lysed extracts were centrifuged at 13,000 g for 10 min at room temperature, and the protein concentration was determined using a commercially available colorimetric assay (BCA Protein Assay Kit). Approximately 25 µg of protein was resolved by 10% or 12,5% SDS-PAGE, and they were then transferred to a PVDF membrane (Hybond-ECL, Amersham Biosciences). The membranes were blocked for 1 h at room temperature in TBS-T (10-mM Tris–HCl, pH 7.5, 100-mM NaCl, and 0.1% Tween-20) with 5% skimmed milk, and then incubated overnight at 4oC with the corresponding primary antibody diluted in TBS-T with 2,5% skimmed milk. After washing 3 times with TBS-T, the membranes were incubated for 2 h at room temperature with their corresponding secondary antibody, diluted in TBS-T with 2,5% skimmed milk. The proteins were visible by enhanced chemiluminescence with ECL (Pierce) using Amersham imager 680, and the signal was quantified by Fiji-ImageJ software.

Proliferative index (Ki67 analysis)

The number of Ki67+ cells with respect to the total number of nuclei labelled with DAPI within a given neurosphere were counted to calculate the Proliferative Index (PI), which was expressed as the % of Ki67+ cells over total DAPI+ cells. More than 20 neurospheres coming from different pictures were analyzed.

Sphere size measurement

Glioblastoma cells (GB5 cell line) were infected by control lentivirus (LV-GFP) or by lentivirus directing expression of *RAP2A* (LV-*RAP2A*). The tumor spheres were Accumax-dissociated to single cells, during three consecutive passages. Seven days after the third plating in a 6-well plate, spheres were captured using a 10x objective (Carl Zeiss Axio Imager A1 microscope). 20 random pictures of each genotype were taken and the images were analyzed by measuring the area of all the spheres present, using the “freeform drawing tool” in the ImageJ software package. The data presented show the average area measured in 3 independent experiments.

Pair assay and Numb segregation analysis

Dissociated cells were plated at low density (50.000 cells/ml) on Matrigel-coated well glass chamber slides on DMEM: F12 with FGF-2 (20 ng/ml), EGF-2 (20 ng/ml) and N2 supplements, 0.1% penicillin (100 U/ml) and streptomycin (100 µg/ml), supplemented with Puromycin. Cultures were incubated for 24, 48 or 72 hours at 37oC with 5% CO_2. Bright field microscopy images from each time point were taken and then the slides were fixed and immunolabeled for Numb, as well as for DAPI to identify nuclei. First, isolated groups of cells (most probably coming from the same single progenitor) were identified and the number of cells in each cluster was annotated. Groups of 3-5 cells were counted in the slide incubated for 48h and groups of 6-8 cells in that incubated for 72h. We did not include in the analysis “neurospheres” with 2 cells, as these can be the result of a symmetric or an asymmetric cell division. We took into account neurospheres with 3-5-7 cells (as odd number of cells) and 4-6-8 cells (as even number of cells).

For the Numb analysis, the localization of Numb was visualized and scored in at least 100 DAPI-labeled cell pairs. To avoid any artefact, pairs were only captured in the slide incubated for 24h. Images in **Figure 6** were acquired in an Inverted Microscope Leica Thunder Imager.

Statistical analysis

The data were first analyzed using the Shapiro-Wilk test to determine whether the sample followed a normal distribution. Parametric T-test or a nonparametric two-tailed Mann Whitney U test for those that did not follow a normal distribution were used to compare statistical differences between two different groups. Mean and standard deviation values were calculated by standard methods. A Chi-square test was used to analyze the data in **Figure 2C**. To assume statistical significance, p-values were determined below 0.05. Sample size (n) and the p-value are indicated in the figure or figure legend; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns: not significant ($p > 0.05$).

Data availability

No big datasets are generated

Acknowledgements

We thank the Bloomington *Drosophila* Stock Center at the University of Indiana and the Developmental Studies Hybridoma Bank at the University of Iowa for kindly providing fly strains and reagents. R.G. was funded by “Instituto de Salud Carlos III (ISCIII)” through the project “Miguel Servet Contracts” (CP21/00116), PI22/01171 and co-funded by the European Union. M.S. was supported by the Spanish grant from the Instituto de Salud Carlos III (ISCIII) PI22/00824. A.C. was financed by the Spanish grants from the Ministry of Science, Innovation and Universities (PGC2018-097093-B-100), the Ministry of Science and Innovation (PID2021-123196NB-100), the Generalitat Valenciana Prometeo/2021/052 grant and by FEDER (European Regional Development Fund). The “Instituto de Neurociencias” in Alicante is a “Severo Ochoa Center of Excellence in R&D”. Our group belongs to “Conexión Cáncer-CSIC”.

Additional information

Author contributions

M.F. participated in the design of the experiments, conducted most of them and analyzed the data; D.B. completed some experiments; R.G. developed the GB lines and performed the in silico analyses; V.M.B. and M.S. carried out the GB microarray and bioinformatics analysis; A.C. conceived the study and wrote the manuscript.

Funding

Spanish Ministry of Science, Innovation and Universitites (PGC2018-097093-B-100)

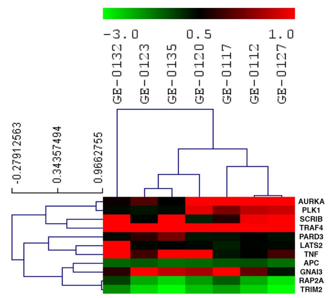
Spanish Ministry of Science and Innovation (PID2021-123196NB-100)

Generalitat Valenciana (Prometeo/2021/052)

European Regional Development Fund

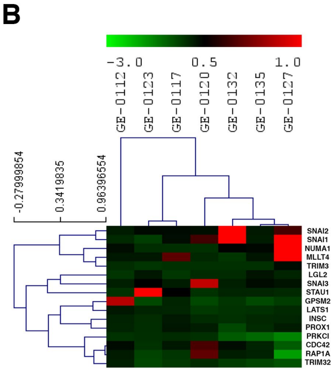
Supplemental figures

A



SIGNIFICANT GENES	Gene mean	Gene standart deviation	T-value	Degrees of freedom	Raw p-value	Adj. P-value	False Discovery Rate
APC	-0,8564163	0,17536853	12,920588	6	1,32E-05	1,32E-05	3,57E-04
AURKA	1,1348615	0,59970766	5,006708	6	0,002436024	0,002436024	0,008221583
GNAI3	0,80726486	0,3571742	5,9797773	6	9,82E-04	9,82E-04	0,006626926
LATS2	0,48613244	0,33597156	3,828257	6	0,008676765	0,008676765	0,023427265
PARD3	0,41569003	0,19338576	5,687143	6	0,001275032	0,001275032	0,006885173
PLK1	0,6329842	0,3034217	5,519443	6	0,001487693	0,001487693	0,006694619
RAP2A	-1,6664003	0,82468957	5,3461094	6	0,001751076	0,001751076	0,006754152
SCRIB	0,86455333	0,5380772	4,2510505	6	0,005373919	0,005373919	0,016121758
TNF	1,0754648	0,75984955	3,7447052	6	0,00956926	0,00956926	0,023488184
TRAF4	1,5386126	0,5544137	7,3425074	6	3,26E-04	3,26E-04	0,002937794
TRIM2	-2,4467607	0,76215523	8,493703	6	1,46E-04	1,46E-04	0,001967374

B

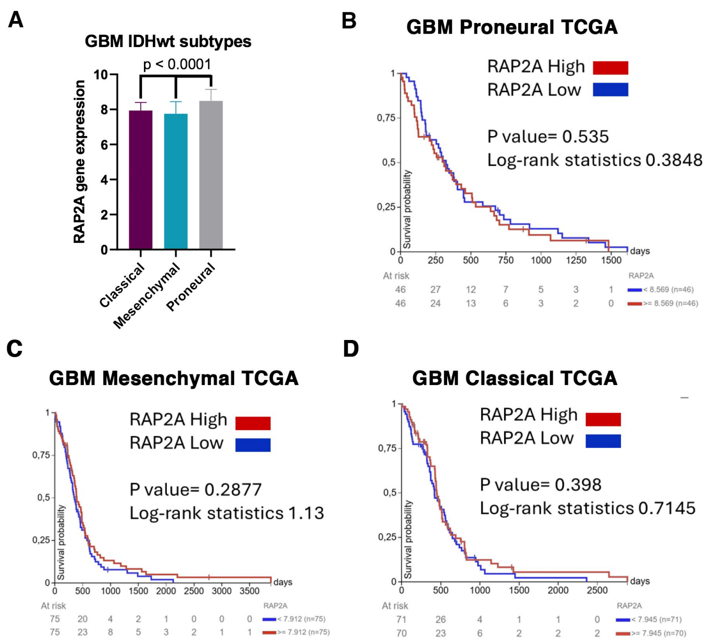


NON-SIGNIFICANT GENES	Gene mean	Gene standart deviation	T-value	Degrees of freedom	Raw p-value	Adj. P-value	False Discovery Rate
CDC42	0,072670326	0,45961553	0,4183227	6	0,6902709	0,6902709	0,6902709
GPSM2	-0,16391191	0,48773852	0,8891447	6	0,40815637	0,40815637	0,50091916
INSC	-0,032859176	0,09984764	0,87069875	6	0,4173893	0,4173893	0,48997873
LATS1	-0,07757898	0,1556407	1,3187724	6	0,23534298	0,23534298	0,33443475
LLGL2	-0,035943888	0,18552625	0,5125883	6	0,6265566	0,6265566	0,6506549
MLLT4	0,32220197	0,5118863	1,665343	6	0,14689733	0,14689733	0,23330753
NUMA1	0,25123188	0,42919657	1,5487008	6	0,17242447	0,17242447	0,25863668
PRKCI	-0,51130617	0,48042583	2,8158123	6	0,03052182	0,03052182	0,06339147
PROX1	-0,044319995	0,22097129	0,53065574	6	0,61471516	0,61471516	0,6638924
RAP1A	-0,19811071	0,7374828	0,71073073	6	0,50392807	0,50392807	0,5669191
SNAI2	0,5057944	0,51646173	2,5911043	6	0,041150834	0,041150834	0,0740715
SNAI3	0,3120758	0,29555163	2,7936742	6	0,03142489	0,03142489	0,06060515
SNAI1	0,61279476	0,87907904	1,8443193	6	0,11468955	0,11468955	0,19353862
STAU1	0,1599102	0,46843663	0,90318006	6	0,40123484	0,40123484	0,5158734
TRIM3	-0,09416514	0,19190228	1,2982521	6	0,2418671	0,2418671	0,3265206
TRIM32	-0,30926165	0,2393221	3,4189463	6	0,014162053	0,014162053	0,03186462

Supplemental Figure 1.

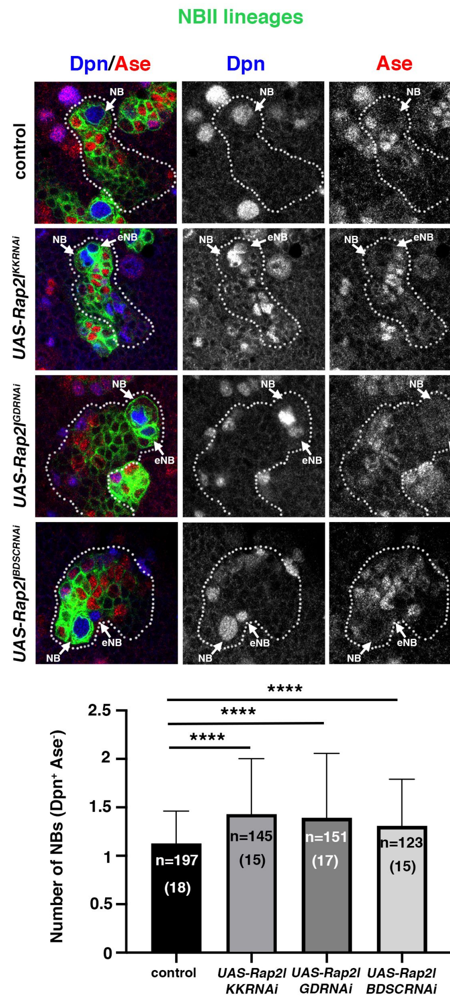
Statistical analysis of asymmetric cell division expression levels in GBM patient samples.

A hierarchical clustering of genes showing significant or non-significant changes in their expression levels is shown. A t-test was applied to the microarray values of the asymmetric cell division regulator genes shown in Fig.1. The statistical values were presented in separate tables corresponding to the significant (**A**) and non-significant (**B**) changes in the indicated genes. The overall threshold *p*-value was 0.01. *p*-values were based on t distribution. Significance was determined using just Alpha.



Supplemental Figure 2.

(A) RAP2A expression levels analysis according to IDHwt GBM subtypes in the TCGA cohort. **(B-D)** Kaplan-Meier survival curves for each GBM subtype, Proneural **(B)**, Mesenchymal **(C)**, and Classical **(D)**, based on high and low RAP2A expression levels.



Supplemental Figure 3.

***Drosophila RAP2A* homolog *Rap2l* regulates ACD.**

Confocal micrographs showing a control larval brain NBII lineage with one NB (Dpn$^+$ Ase$^-$) and NBII lineages in which *Rap2l* has been downregulated displaying an ectopic NB (eNB). The phenotype of three different, independent *Rap2l*^{RNAi} lines (KK, GD and BDSC) are shown. Data represented in the bar graph was analyzed with a U Mann-Whitney test; ****$p<0.0001$; n=number of NB lineages analyzed; the number of different brains analyzed per genotype is also indicated in brackets.

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Reviewer #1 (Public review):

Summary:

The authors validate the contribution of RAP2A to GB progression. RAp2A participates in asymetric cell division, and the localization of several cell polarity markers including cno and Numb.

Strengths:

The use of human data, Drosophila models and cell culture or neurospheres is a good scenario to validate the hypothesis using complementary systems.

Moreover, the mechanisms that determine GB progression, and in particular glioma stem cells biology, are relevant for the knowledge on glioblastoma and opens new possibilities to future clinical strategies.

Weaknesses:

While the manuscript presents a well-supported investigation into RAP2A's role in GBM, some methodological aspects could benefit from further validation. The major concern is the reliance on a single GB cell line (GB5), including multiple GBM lines, particularly primary patient-derived 3D cultures with known stem-like properties, would significantly enhance the study's robustness.

Several specific points raised in previous reviews have improved this version of the manuscript:

• The specificity of Rap2l RNAi has been further confirmed by using several different RNAi tools.

• Quantification of phenotypic penetrance and survival rates in Rap2l mutants would help determine the consistency of ACD defects. The authors have substantially increased the number of samples analyzed including three different RNAi lines (both the number of NB lineages and the number of different brains analyzed) to confirm the high penetrance of the phenotype.

• The observations on neurosphere size and Ki-67 expression require normalization (e.g., Ki-67+ cells per total cell number or per neurosphere size). This is included in the manuscript and now clarified in the text.

• The discrepancy in Figures 6A and 6B requires further discussion. The authors have included a new analysis and further explanations and they can conclude that in 2 cell-neurospheres there are more cases of asymmetric divisions in the experimental condition (RAP2A) than in the control.

• Live imaging of ACD events would provide more direct evidence. Live imaging was not done due to technical limitations. Despite being a potential contribution to the manuscript, the current conclusions of the manuscript are supported by the current data, and live experiments can be dispensable

• Clarification of terminology and statistical markers (e.g., p-values) in Figure 1A would improve clarity. This has been improved.

Comments on revisions:

The manuscript has improved the clarity in general, and I think that it is suitable for publication. However, for future experiments and projects, I would like to insist in the relevance of validating the results in vivo using xenografts with 3D-primary patient-derived cell lines or GB organoids.

https://doi.org/10.7554/eLife.105690.2.sa2

Reviewer #2 (Public review):

This study investigates the role of RAP2A in regulating asymmetric cell division (ACD) in glioblastoma stem cells (GSCs), bridging insights from Drosophila ACD mechanisms to human tumor biology. They focus on RAP2A, a human homolog of Drosophila Rap2l, as a novel ACD regulator in GBM is innovative, given its underexplored role in cancer stem cells (CSCs). The hypothesis that ACD imbalance (favoring symmetric divisions) drives GSC expansion and tumor progression introduces a fresh perspective on differentiation therapy. However, the dual role of ACD in tumor heterogeneity (potentially aiding therapy resistance) requires deeper discussion to clarify the study's unique contributions against existing controversies.

Comments on revisions:

More experiments as suggested in the original assessment of the submission are needed to justify the hypothesis drawn in the manuscript.

https://doi.org/10.7554/eLife.105690.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The authors validate the contribution of RAP2A to GB progression. RAp2A participates in asymmetric cell division, and the localization of several cell polarity markers, including cno and Numb.*
>
> *Strengths:*
>
> *The use of human data, Drosophila models, and cell culture or neurospheres is a good scenario to validate the hypothesis using complementary systems.*
>
> *Moreover, the mechanisms that determine GB progression, and in particular glioma stem cells biology, are relevant for the knowledge on glioblastoma and opens new possibilities to future clinical strategies.*
>
> *Weaknesses:*
>
> *While the manuscript presents a well-supported investigation into RAP2A's role in GBM, several methodological aspects require further validation. The major concern is the reliance on a single GB cell line (GB5), which limits the generalizability of the findings. Including multiple GBM lines, particularly primary patient-derived 3D cultures with known stem-like properties, would significantly enhance the study's relevance.*
>
> *Additionally, key mechanistic aspects remain underexplored. Further investigation into the conservation of the Rap2l-Cno/aPKC pathway in human cells through rescue experiments or protein interaction assays would be beneficial. Similarly, live imaging or lineage tracing would provide more direct evidence of ACD frequency, complementing the current indirect metrics (odd/even cell clusters, Numb asymmetry).*
>
> *Several specific points require attention:*
>
> *(1) The specificity of Rap2l RNAi needs further confirmation. Is Rap2l expressed in neuroblasts or intermediate neural progenitors? Can alternative validation methods be*

> *employed?*

There are no available antibodies/tools to determine whether Rap2l is expressed in NB lineages, and we have not been able either to develop any. However, to further prove the specificity of the Rap2l phenotype, we have now analyzed two additional and independent RNAi lines of Rap2l along with the original RNAi line analyzed. We have validated the results observed with this line and found a similar phenotype in the two additional RNAi lines now analyzed. These results have been added to the text ("Results section", page 6, lines 142-148) and are shown in Supplementary Figure 3.

> *(2) Quantification of phenotypic penetrance and survival rates in Rap2l mutants would help determine the consistency of ACD defects.*

In the experiment previously mentioned (repetition of the original Rap2l RNAi line analysis along with two additional Rap2l RNAi lines) we have substantially increased the number of samples analyzed (both the number of NB lineages and the number of different brains analyzed). With that, we have been able to determine that the penetrance of the phenotype was 100% or almost 100% in the 3 different RNAi lines analyzed (n>14 different brains/larvae analyzed in all cases). Details are shown in the text (page 6, lines 142-148), in Supplementary Figure 3 and in the corresponding figure legend.

> *(3) The observations on neurosphere size and Ki-67 expression require normalization (e.g., Ki-67+ cells per total cell number or per neurosphere size). Additionally, apoptosis should be assessed using Annexin V or TUNEL assays.*

The experiment of Ki-67+ cells was done considering the % of Ki-67+ cells respect the total cell number in each neurosphere. In the "Materials and methods" section it is well indicated: "The number of Ki67+ cells with respect to the total number of nuclei labelled with DAPI within a given neurosphere were counted to calculate the Proliferative Index (PI), which was expressed as the % of Ki67+ cells over total DAPI+ cells"

Perhaps it was not clearly showed in the graph of Figure 5A. We have now changed it indicating: "% of Ki67+ cells/ neurosphere" in the "Y axis".

Unfortunately, we currently cannot carry out neurosphere cultures to address the apoptosis experiments.

> *(4) The discrepancy in Figures 6A and 6B requires further discussion.*

We agree that those pictures can lead to confusion. In the analysis of the "% of neurospheres with even or odd number of cells", we included the neurospheres with 2 cells both in the control and in the experimental condition (RAP2A). The number of this "2 cell-neurospheres" was very similar in both conditions (27,7 % and 27 % of the total neurospheres analyzed in each condition), and they can be the result of a previous symmetric or asymmetric division, we cannot distinguish that (only when they are stained with Numb, for example, as shown in Figure 6B). As a consequence, in both the control and in the experimental condition, these 2-cell neurospheres included in the group of "even" (Figure 6A) can represent symmetric or asymmetric divisions. However, in the experiment shown in Figure 6B, it is shown that in these 2 cellneurospheres there are more cases of asymmetric divisions in the experimental condition (RAP2A) than in the control.

Nevertheless, to make more accurate and clearer the conclusions, we have reanalyzed the data taking into account only the neurospheres with 3-5-7 (as odd) or 4-6-8 (as even) cells. Likewise, we have now added further clarifications regarding the way the experiment has been analyzed in the methods.

> *(5) Live imaging of ACD events would provide more direct evidence.*

We agree that live imaging would provide further evidence. Unfortunately, we currently cannot carry out neurosphere cultures to approach those experiments.

> *(6) Clarification of terminology and statistical markers (e.g., p-values) in Figure 1A would improve clarity.*

We thank the reviewer for pointing out this issue. To improve clarity, we have now included a Supplementary Figure (Fig. S1) with the statistical parameters used. Additionally, we have performed a hierarchical clustering of genes showing significant or not-significant changes in their expression levels.

> *(7) Given the group's expertise, an alternative to mouse xenografts could be a Drosophila genetic model of glioblastoma, which would provide an in vivo validation system aligned with their research approach.*

The established Drosophila genetic model of glioblastoma is an excellent model system to get deep insight into different aspects of human GBM. However, the main aim of our study was to determine whether an imbalance in the mode of stem cell division, favoring symmetric divisions, could contribute to the expansion of the tumor. We chose human GBM cell lines-derived neurospheres because in human GBM it has been demonstrated the existence of cancer stem cells (glioblastoma or glioma stem cells -GSCs--). And these GSCs, as all stem cells, can divide symmetric or asymmetrically. In the case of the Drosophila model of GBM, the neoplastic transformation observed after overexpressing the EGF receptor and PI3K signaling is due to the activation of downstream genes that promote cell cycle progression and inhibit cell cycle exit. It has also been suggested that the neoplastic cells in this model come from committed glial progenitors, not from stem-like cells.

With all, it would be difficult to conclude the causes of the potential effects of manipulating the Rap2l levels in this Drosophila system of GBM. We do not discard this analysis in the future (we have all the "set up" in the lab). However, this would probably imply a new project to comprehensively analyze and understand the mechanism by which Rap2l (and other ACD regulators) might be acting in this context, if it is having any effect.

However, as we mentioned in the Discussion, we agree that the results we have obtained in this study must be definitely validated in vivo in the future using xenografts with 3D-primary patient-derived cell lines.

> ***Reviewer #2 (Public review):***
>
> *This study investigates the role of RAP2A in regulating asymmetric cell division (ACD) in glioblastoma stem cells (GSCs), bridging insights from Drosophila ACD mechanisms to human tumor biology. They focus on RAP2A, a human homolog of Drosophila Rap2l, as a novel ACD regulator in GBM is innovative, given its underexplored role in cancer stem cells (CSCs). The hypothesis that ACD imbalance (favoring symmetric divisions) drives GSC expansion and tumor progression introduces a fresh perspective on differentiation therapy. However, the dual role of ACD in tumor heterogeneity (potentially aiding therapy resistance) requires deeper discussion to clarify the study's unique contributions against existing controversies. Some limitations and questions need to be addressed.*
>
> *(1) Validation of RAP2A's prognostic relevance using TCGA and Gravendeel cohorts strengthens clinical relevance. However, differential expression analysis across GBM subtypes (e.g., MES, DNA-methylation subtypes) should be included to confirm specificity.*

We have now included a Supplementary figure (Supplementary Figure 2), in which we show the analysis of RAP2A levels in the different GBM subtypes (proneural, mesenchymal and classical) and their prognostic relevance (i.e. the proneural subtype that presents RAP2A levels significantly higher than the others is the subtype that also shows better prognostic).

> *(2) Rap2l knockdown-induced ACD defects (e.g., mislocalization of Cno/Numb) are well-designed. However, phenotypic penetrance and survival rates of Rap2l mutants should be quantified to confirm consistency.*

We have now analyzed two additional and independent RNAi lines of Rap2l along with the original RNAi line. We have validated the results observed with this line and found a similar phenotype in the two additional RNAi lines now analyzed. To determine the phenotypic penetrance, we have substantially increased the number of samples analyzed (both the number of NB lineages and the number of different brains analyzed). With that, we have been able to determine that the penetrance of the phenotype was 100% or almost 100% in the 3 different Rap2l RNAi lines analyzed (n>14 different brains/larvae analyzed in all cases). These results have been added to the text ("Results section", page 6, lines 142-148) and are shown in Supplementary Figure 3 and in the corresponding figure legend.

> *(3) While GB5 cells were effectively used, justification for selecting this line (e.g., representativeness of GBM heterogeneity) is needed. Experiments in additional GBM lines (especially the addition of 3D primary patient-derived cell lines with known stem cell phenotype) would enhance generalizability.*

We tried to explain this point in the paper (Results). As we mentioned, we tested six different GBM cell lines finding similar mRNA levels of RAP2A in all of them, and significantly lower levels than in control Astros (Fig. 3A). We decided to focus on the GBM cell line called GB5 as it grew well (better than the others) in neurosphere cell culture conditions, for further analyses. We agree that the addition of at least some of the analyses performed with the GB5 line using other lines (ideally in primary patientderive cell lines, as the reviewer mentions) would reinforce the results. Unfortunately, we cannot perform experiments in cell lines in the lab currently. We will consider all of this for future experiments.

> *(4) Indirect metrics (odd/even cell clusters, NUMB asymmetry) are suggestive but insufficient. Live imaging or lineage tracing would directly validate ACD frequency.*

We agree that live imaging would provide further evidence. Unfortunately, we cannot approach those experiments in the lab currently.

> *(5) The initial microarray (n=7 GBM patients) is underpowered. While TCGA data mitigate this, the limitations of small cohorts should be explicitly addressed and need to be discussed.*

We completely agree with this comment. We had available the microarray, so we used it as a first approach, just out of curiosity of knowing whether (and how) the levels of expression of those human homologs of Drosophila ACD regulators were affected in this small sample, just as starting point of the study. We were conscious of the limitations of this analysis and that is why we followed up the analysis in the datasets, on a bigger scale. We already mentioned the limitations of the array in the Discussion:

"The microarray we interrogated with GBM patient samples had some limitations. For example, not all the human genes homologs of the Drosophila ACD regulators were present (i.e. the human homologs of the determinant Numb). Likewise, we only tested seven different GBM patient samples. Nevertheless, the output from this analysis was enough to determine that most of the human genes tested in the array presented altered levels of expression"[....] In silico analyses, taking advantage of the existence of established datasets, such as the TCGA, can help to more robustly assess, in a bigger sample size, the relevance of those human genes expression levels in GBM progression, as we observed for the gene RAP2A."

> *(6) Conclusions rely heavily on neurosphere models. Xenograft experiments or patient-derived orthotopic models are critical to support translational relevance, and such basic research work needs to be included in journals.*

We completely agree. As we already mentioned in the Discussion, the results we have obtained in this study must be definitely validated in vivo in the future using xenografts with 3D-primary patient-derived cell lines.

> *(7) How does RAP2A regulate NUMB asymmetry? Is the Drosophila Rap2l-Cno/aPKC pathway conserved? Rescue experiments (e.g., Cno/aPKC knockdown with RAP2A overexpression) or interaction assays (e.g., Co-IP) are needed to establish molecular mechanisms.*

The mechanism by which RAP2A is regulating ACD is beyond the scope of this paper. We do not even know how Rap2l is acting in Drosophila to regulate ACD. In past years, we did analyze the function of another Drosophila small GTPase, Rap1 (homolog to human RAP1A) in ACD, and we determined the mechanism by which Rap1 was regulating ACD (including the localization of Numb): interacting physically with Cno and other small GTPases, such as Ral proteins, and in a complex with additional ACD regulators of the "apical complex" (aPKC and Par-6). Rap2l could be also interacting physically with the "Ras-association" domain of Cno (domain that binds small GTPases, such as Ras and Rap1). We have added some speculations regarding this subject in the Discussion:

"It would be of great interest in the future to determine the specific mechanism by which Rap2l/RAP2A is regulating this process. One possibility is that, as it occurs in the case of the Drosophila ACD regulator Rap1, Rap2l/RAP2A is physically interacting or in a complex with other relevant ACD modulators."

> *(8) Reduced stemness markers (CD133/SOX2/NESTIN) and proliferation (Ki-67) align with increased ACD. However, alternative explanations (e.g., differentiation or apoptosis) must be ruled out via GFAP/Tuj1 staining or Annexin V assays.*

We agree with these possibilities. Regarding differentiation, the potential presence of increased differentiation markers would be in fact a logic consequence of an increase in ACD divisions/reduced stemness markers. Unfortunately, we cannot approach those experiments in the lab currently.

> *(9) The link between low RAP2A and poor prognosis should be validated in multivariate analyses to exclude confounding factors (e.g., age, treatment history).*

We have now added this information in the "Results section" (page 5, lines 114-123).

> *(10) The broader ACD regulatory network in GBM (e.g., roles of other homologs like NUMB) and potential synergies/independence from known suppressors (e.g., TRIM3) warrant exploration.*

The present study was designed as a "proof-of-concept" study to start analyzing the hypothesis that the expression levels of human homologs of known Drosophila ACD regulators might be relevant in human cancers that contain cancer stem cells, if those human homologs were also involved in modulating the mode of (cancer) stem cell division.

To extend the findings of this work to the whole ACD regulatory network would be the logic and ideal path to follow in the future.

We already mentioned this point in the Discussion:

"....it would be interesting to analyze in the future the potential consequences that altered levels of expression of the other human homologs in the array can have in the behavior of the GSCs. In silico analyses, taking advantage of the existence of established datasets, such as the TCGA, can help to more robustly assess, in a bigger sample size, the relevance of those human genes expression levels in GBM progression, as we observed for the gene RAP2A."

> *(11) The figures should be improved. Statistical significance markers (e.g., p-values) should be added to Figure 1A; timepoints/culture conditions should be clarified for Figure 6A.*

Regarding the statistical significance markers, we have now included a Supplementary Figure (Fig. S1) with the statistical parameters used. Additionally, we have performed a hierarchical clustering of genes showing significant or notsignificant changes in their expression levels.

Regarding the experimental conditions corresponding to Figure 6A, those have now been added in more detail in "Materials and Methods" ("Pair assay and Numb segregation analysis" paragraph).

> *(12) Redundant Drosophila background in the Discussion should be condensed; terminology should be unified (e.g., "neurosphere" vs. "cell cluster").*

As we did not mention much about Drosophila ACD and NBs in the "Introduction", we needed to explain in the "Discussion" at least some very basic concepts and information about this, especially for "non-drosophilists". We have reviewed the Discussion to maintain this information to the minimum necessary.

We have also reviewed the terminology that the Reviewer mentions and have unified it.

> ***Reviewer #1 (Recommendations for the authors):***
>
> *To improve the manuscript's impact and quality, I would recommend:*
>
> *(1) Expand Cell Line Validation: Include additional GBM cell lines, particularly primary patient-derived 3D cultures, to increase the robustness of the findings.*
>
> *(2) Mechanistic Exploration: Further examine the conservation of the Rap2l/Cno/aPKC pathway in human cells using rescue experiments or protein interaction assays.*
>
> *(3) Direct Evidence of ACD: Implement live imaging or lineage tracing approaches to strengthen conclusions on ACD frequency.*
>
> *(4) RNAi Specificity Validation: Clarify Rap2l RNAi specificity and its expression in neuroblasts or intermediate neural progenitors.*
>
> *(5) Quantitative Analysis: Improve quantification of neurosphere size, Ki-67 expression, and apoptosis to normalize findings.*
>
> *(6) Figure Clarifications: Address inconsistencies in Figures 6A and 6B and refine statistical markers in Figure 1A.*

> *(7) Alternative In Vivo Model: Consider leveraging a Drosophila glioblastoma model as a complementary in vivo validation approach.*
>
> *Addressing these points will significantly enhance the manuscript's translational relevance and overall contribution to the field.*

We have been able to address points 4, 5 and 6. Others are either out of the scope of this work (2) or we do not have the possibility to carry them out at this moment in the lab (1, 3 and 7). However, we will complete these requests/recommendations in other future investigations.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *Major Revision /insufficient required to address methodological and mechanistic gaps.*
>
> *(1) Enhance Clinical Relevance*
>
> *Validate RAP2A's prognostic significance across multiple GBM subtypes (e.g., MES, DNA-methylation subtypes) using datasets like TCGA and Gravendeel to confirm specificity.*
>
> *Perform multivariate survival analyses to rule out confounding factors (e.g., patient age, treatment history).*
>
> *(2) Strengthen Mechanistic Insights*
>
> *Investigate whether the Rap2l-Cno/aPKC pathway is conserved in human GBM through rescue experiments (e.g., RAP2A overexpression with Cno/aPKC knockdown) or interaction assays (e.g., Co-IP).*
>
> *Use live-cell imaging or lineage tracing to directly validate ACD frequency instead of relying on indirect metrics (odd/even cell clusters, NUMB asymmetry).*
>
> *(3) Improve Model Systems & Experimental Design*
>
> *Justify the selection of GB5 cells and include additional GBM cell lines, particularly 3D primary patient-derived cell models, to enhance generalizability.*
>
> *It is essential to perform xenograft or orthotopic patient-derived models to support translational relevance.*
>
> *(5) Address Alternative Interpretations*
>
> *Rule out other potential effects of RAP2A knockdown (e.g., differentiation or apoptosis) using GFAP/Tuj1 staining or Annexin V assays.*
>
> *Explore the broader ACD regulatory network in GBM, including interactions with NUMB and TRIM3, to contextualize findings within known tumor-suppressive pathways.*
>
> *(6) Improve Figures & Clarity*
>
> *Add statistical significance markers (e.g., p-values) in Figure 1A and clarify timepoints/culture conditions for Figure 6A.*
>
> *Condense redundant Drosophila background in the discussion and ensure consistent terminology (e.g., "neurosphere" vs. "cell cluster").*

We have been able to address points 1, partially 3 and 6. Others are either out of the scope of this work or we do not have the possibility to carry them out at this moment in the lab.

However, we are very interested in completing these requests/recommendations and we will approach that type of experiments in other future investigations.

https://doi.org/10.7554/eLife.105690.2.sa0